

Code:

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#include<iostream>
#include<iomanip>
#include<cmath>
using namespace std;
int main()
{
    cout.precision(4);
    cout.setf(ios::fixed);
    int n,i,j;
    cout<<"\nEnter the order of the matrix:\n";
    cin>>n;
    double a[n][n],b[n],c[n],k,eps,y;

    cout<<"\nEnter the elements of matrix row-wise:\n";
    for (i=0;i<n;i++)          //Get the elements of the matrix
        for (j=0;j<n;j++)
            cin>>a[i][j];

    cout<<"\nEnter the initial guess for the eigen-vector:\n";
    for (j=0;j<n;j++)          //Get the initial guess for the eigen vector
        cin>>b[j];

    cout<<"\nEnter the accuracy desired:\n";
    cin>>eps;
    k=b[0];                    //Assign some initial value to the eigen value, 'k'
    do
    {
        y=k;
        for (i=0;i<n;i++)
        {
            c[i]=0;
            for (j=0;j<n;j++)
                c[i]=c[i]+a[i][j]*b[j];    //After all the iterations axb=c
        }
        k=c[0];
        for (i=1;i<n;i++)
        {
            if(abs(k)<abs(c[i]))
                k=c[i];
        }
        for (i=0;i<n;i++)
            b[i]=c[i]/k;    //Calculate the new Eigen Vector
    }while(abs(k-y)>=eps);

    cout<<"\n\nThe largest Eigenvalue is: "<<k<<endl;
    cout<<"\nAnd the corresponding Eigenvector is: \n";
    for (i=0;i<n;i++)
        cout<<b[i]<<endl;
    cout<<"\n";
    return 0;
}

```

Output:

Enter the order of the matrix:

2

Enter the elements of matrix row-wise:

4 1

2 3

Enter the initial guess for the eigen-vector:

1

0

Enter the accuracy desired:

0.01

The largest Eigenvalue is: 4.9939

And the corresponding Eigenvector is:

1.0000

0.9975

Process exited after 66.05 seconds with return value 0

Press any key to continue . . .